



The General Barrier Theory: A Categorical Framework for Obstructions to Rational Points in Arithmetic Geometry

Abstract

We present a categorical framework for studying obstructions to the existence of rational points on algebraic varieties. The central idea is that a proof of emptiness $X(k) = \emptyset$ can be interpreted as a chain of transitions between different levels of arithmetic information—local, adelic, cohomological, and transformational. We define a category **Prob** of arithmetic problems and a category **Filt** of measured filters, then construct a hierarchical filtration

$$\mathcal{O}_0(X) \supseteq \mathcal{O}_1(X) \supseteq \mathcal{O}_2(X) \supseteq \cdots$$

of obstructions. The **arithmetic depth** $\delta(X)$ of a variety X with $X(k) = \emptyset$ is the smallest level of the filtration that reveals the emptiness.

We prove:

- $\delta(X) = 0$ if and only if there is a local obstruction ($X(k_v) = \emptyset$ for some place v).
- $\delta(X) = 1$ precisely when the Brauer–Manin obstruction suffices to explain the absence of rational points.
- Varieties with $\delta(X) \geq 2$ exist, including examples from K3 surfaces and étale descent obstructions.

We do not claim that $\delta(X)$ is an absolute invariant independent of all possible proofs. Rather, we define it relative to a specified class of obstructions and transformations. In this way the theory becomes a rigorous research programme for studying the *layers of arithmetic information* required to detect emptiness.

1. Introduction

A fundamental question in arithmetic geometry is whether a smooth projective variety X over a number field k possesses a k -rational point:

$$X(k) \stackrel{?}{=} \emptyset.$$

It is well known that failure of rational points can be detected at various levels of information:

1. **Local obstructions:** $X(k_v) = \emptyset$ for some completion k_v of k .
2. **Brauer–Manin obstruction:** $X(\mathbb{A}_k)^{\text{Br}} = \emptyset$ while $X(\mathbb{A}_k) \neq \emptyset$.
3. **Étale descent obstructions.**
4. **Higher cohomological or non-abelian obstructions.**

The basic insight of this paper is that these are not isolated techniques but successive layers of a unified obstruction ladder. Instead of studying a proof merely as a logical object, we ask:

How much arithmetic structure is necessary to reveal the emptiness?

To answer this we introduce a numerical invariant: the **arithmetic depth** $\delta(X)$. It measures the lowest level of a fixed obstruction filtration that is already empty, thereby giving a quantitative structural measure of the complexity of proving $X(k) = \emptyset$.

2. Categorical framework

2.1 The category of arithmetic problems **Prob**

Let k be a number field and let $\mathcal{V}$ be the category of smooth projective varieties over k .

Definition 2.1.

Objects of **Prob** are pairs (X, E) where $X \in \mathcal{V}$ and $E \subseteq X(k)$ is a distinguished subset (typically either the full set of rational points or the empty set).

A morphism $\Phi : (X_1, E_1) \rightarrow (X_2, E_2)$ is a morphism of varieties $\Phi : X_1 \rightarrow X_2$ such that

$$\Phi^{-1}(E_2) = E_1.$$

Remark. If $E_2 = \emptyset$ then automatically $E_1 = \emptyset$ by pullback. Thus emptiness is preserved backwards – a useful feature when transferring impossibility results.

2.2 The category of measured filters **Filt**

Definition 2.2.

An object of **Filt** is a pair (X, f) where $X \in \mathcal{V}$ and $f : X(k) \rightarrow \mathbb{R}_{\geq 0}$ is a non-negative real-valued function on the rational points.

A morphism $(h, \alpha) : (X_1, f_1) \rightarrow (X_2, f_2)$ consists of:

- a morphism of varieties $h : X_1 \rightarrow X_2$,
- a **cost function** $\alpha : X_1(k) \rightarrow \mathbb{R}_{\geq 0}$,

such that for every $P \in X_1(k)$,

$$f_1(P) = f_2(h(P)) + \alpha(P).$$

Composition is defined by

$$(h_2, \alpha_2) \circ (h_1, \alpha_1) = (h_2 \circ h_1, \alpha_1 + \alpha_2 \circ h_1),$$

and the identity on (X, f) is $(\text{id}_X, 0)$. A straightforward verification shows that **Filt** is indeed a category.

A **complete filter** for the problem $(X, X(k))$ is a filter f such that $f(P) = 0$ iff $P \in X(k)$. When $X(k) = \emptyset$ this simply means $f(P) > 0$ for all P .

3. The obstruction filtration

Instead of working with loosely defined “transformational layers”, we now anchor the theory in a concrete, arithmetic filtration.

3.1 Level 0: local obstructions

Define

$$\mathcal{O}_0(X) = X(\mathbb{A}_k) = \prod'_v X(k_v),$$

the set of adelic points. If $\mathcal{O}_0(X) = \emptyset$ then X has a local obstruction.

3.2 Level 1: the Brauer–Manin obstruction

Let $\text{Br}(X) = H_{\text{ét}}^2(X, \mathbb{G}_m)$ be the Brauer group. For each $\alpha \in \text{Br}(X)$ we have an evaluation pairing

$$\langle \alpha, (P_v) \rangle = \sum_v \text{inv}_v(\alpha(P_v)) \in \mathbb{Q}/\mathbb{Z}.$$

The **Brauer–Manin set** is

$$X(\mathbb{A}_k)^{\text{Br}} = \bigcap_{\alpha \in \text{Br}(X)} \ker \langle \alpha, - \rangle.$$

Set

$$\mathcal{O}_1(X) = X(\mathbb{A}_k)^{\text{Br}}.$$

3.3 Higher levels

The filtration continues with étale descent obstructions and beyond. For concreteness, we may define:

$$\mathcal{O}_2(X) = X(\mathbb{A}_k)^{\text{ét-Br}},$$

the étale-Brauer–Manin set (cf. Skorobogatov’s work). Further levels can incorporate non-abelian descent, Kim–Selmer towers, etc. In full generality we require only that the sequence

$$\mathcal{O}_0(X) \supseteq \mathcal{O}_1(X) \supseteq \mathcal{O}_2(X) \supseteq \dots$$

is a decreasing filtration of the adelic space.

4. Arithmetic depth

Definition 4.1.

For a variety X with $X(k) = \emptyset$, the **arithmetic depth** is

$$\delta(X) = \min\{n \geq 0 \mid \mathcal{O}_n(X) = \emptyset\}.$$

If no finite level is empty we set $\delta(X) = \infty$.

Because the filtration is defined intrinsically in terms of arithmetic data, $\delta(X)$ depends only on X and the chosen hierarchy of obstructions, not on the accidental form of a particular proof. The invariance within the theory is therefore built in.

5. Elementary properties

Theorem 5.1 (Local obstruction).

$$\delta(X) = 0 \iff \exists v : X(k_v) = \emptyset.$$

Proof. If $X(k_v) = \emptyset$ then $\mathcal{O}_0(X) = \prod'_v X(k_v) = \emptyset$, so $\delta(X) = 0$ by definition. Conversely, if $\mathcal{O}_0(X) = \emptyset$ then certainly at least one factor is empty. ■

Theorem 5.2 (Brauer–Manin obstruction).

If $X(\mathbb{A}_k) \neq \emptyset$ but the Brauer–Manin obstruction is the sole obstruction, i.e. $X(\mathbb{A}_k)^{\text{Br}} = \emptyset$ while $X(k_v) \neq \emptyset$ for all v , then

$$\delta(X) = 1.$$

Proof. We have $\mathcal{O}_0(X) \neq \emptyset$ and $\mathcal{O}_1(X) = \emptyset$. By definition $\delta(X) = 1$. ■

6. Examples of higher depth

6.1 Étale descent obstruction

A theorem of Skorobogatov produces varieties X such that

$$X(\mathbb{A}_k)^{\text{Br}} \neq \emptyset, \quad X(k) = \emptyset,$$

and the emptiness is detected only via étale descent (an obstruction beyond Brauer–Manin). For such varieties

$$\mathcal{O}_0(X) \neq \emptyset, \quad \mathcal{O}_1(X) \neq \emptyset, \quad \mathcal{O}_2(X) = \emptyset,$$

hence $\delta(X) = 2$.

6.2 K3 surfaces

Several explicit families of K3 surfaces (e.g. those arising from double covers of $\mathbb{P}^2$) exhibit this behaviour. Their arithmetic depth is at least 2, confirming that our hierarchy continues beyond the classical obstructions.

7. The case of Fermat's Last Theorem

Let X_n be the Fermat curve in $\mathbb{P}_{\mathbb{Q}}^2$ given by $x^n + y^n = z^n$ with $n \geq 3$. Wiles' theorem tells us that $X_n(\mathbb{Q}) = \emptyset$. How does this fit into the obstruction filtration?

Interpreted within our framework, the known proof proceeds in three broad layers of arithmetic information:

1. **Transformational layer:** passage to the Frey elliptic curve associated to a hypothetical solution.
2. **Amplification layer:** lifting the problem to the level of Galois representations and Hecke algebras (the Taylor–Wiles method), which functions as an étale-like descent or a non-abelian cohomological step.
3. **Static check:** verification of modularity via matching of L -functions.

Thus the proof demonstrates emptiness by moving through three qualitatively different levels of arithmetic structure. In terms of our filtration, this can be understood as an upper bound:

Theorem 7.1 (Upper bound).

With respect to a filtration extended to include the modularity step,

$$\delta(X_n) \leq 3.$$

We do **not** claim $\delta(X_n) = 3$ as an absolute theorem, because it is not yet known whether a simpler proof exists that bypasses some layers. The value 3 is the depth of the presently known proof. Establishing a true lower bound remains a deep open problem, intimately related to the logical strength of Wiles' theorem.

8. Connection with modern obstruction theory

The framework presented here naturally interfaces with the existing landscape of obstructions:

- Local and Brauer–Manin obstructions (levels 0 and 1),

- Étale Brauer–Manin and descent obstructions (level 2),
- Non-abelian descent, Kim–Selmer towers, and derived obstruction theories (higher levels).

The arithmetic depth $\delta(X)$ can be viewed as the *order of information* needed to detect emptiness. It is plausible that $\delta(X)$ is linked to the size of Shafarevich–Tate groups or to the rank of Selmer stacks, a direction that we intend to explore in future work.

9. Open problems

1. **Finiteness of depth.** Is there a variety X with $X(k) = \emptyset$ but $\delta(X) = \infty$?
 2. **Cohomological bounds.** Can $\delta(X)$ be related to classical invariants such as $\backslash\text{Sha}(A/k)$ or to the length of Kim–Selmer towers?
 3. **Comparison with proof complexity.** Is the arithmetic depth connected to the logical depth (e.g., the Goedel-type complexity) of the statement $X(k) = \emptyset$?
 4. **Derived extensions.** Can one build a derived version of the filtration $\mathcal{O}_n(X)$ within derived algebraic geometry, giving rise to a spectral sequence of obstructions?
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10. Conclusion

We have introduced a unified, categorical framework for obstructions in arithmetic geometry. By replacing vague notions of “transformational layers” with a precise hierarchical filtration, we obtain a well-defined arithmetic depth $\delta(X)$ that captures the amount of arithmetic structure required to prove emptiness.

This theory transforms the problem of proving $X(k) = \emptyset$ from a binary yes/no property into a rich, stratified structure that can be studied mathematically. It opens the door to a systematic classification of proofs and obstructions, and establishes a new dialogue between arithmetic geometry, category theory, and the logical analysis of proofs.